

TYPED CONTROL, INFORMATION, AND SADDLE ENVELOPES FOR KINETIC COTANGENT PHASES

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ABSTRACT. Exact noiseless observations are retained through coarea posterior currents on their level sets rather than being forced into an interior density class. The B2 measure-valued trace semigroup propagates these beliefs and their collision traces. Prediction followed by observed disintegration defines a Feller belief kernel, while a uniform coarea ratio theorem gives quenched block pressures on strategy-reachable compacta. Exponential contraction of connected hierarchy coordinates proves asymptotic sufficiency of the resolved kinetic state. Continuous-prior Chernoff bounds use an explicit local denominator; LAN remains centered at the exact finite-volume mean.

Platform identifier: HSBG-CONTROL-v2. **Controlling revision:**
ROUND7-REFEREE-POSITIVE-CLOSURE.

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1. ROUND-SEVEN RECONSTRUCTION: COAREA POSTERIOR CURRENTS, QUENCHED BLOCK PRESSURES, AND BELIEF-STATE SUFFICIENCY

This section replaces the round-six controlling module. Exact noiseless observations are retained. Their posteriors are not forced into an interior L^1

Date: August 31, 2026.

2020 *Mathematics Subject Classification.* Primary 49L25, 93E20; Secondary 60G35, 82C40.

Key words and phrases. risk-sensitive control, kinetic equation, filtering, saddle envelope, deterministic information.

density class; they are represented by finite coarea currents in the measure-valued hierarchy of B2. Positive-density observation channels are a special case. Strategy-uniform block limits are proved by a ratio theorem on reachable posterior-current compacta.

1.1. Admissible observation channels and posterior currents. On a finite hard-sphere level let the block observation be

$$Y_k = O_k^u(\Phi_{[t_k, t_{k+1}]} Z_{t_k}),$$

where u is the relaxed control. An admissible noiseless channel is a C^2 submersion on every regular chronological cell, with Jacobian $J_O \geq j_* > 0$ away from a B2 superexponentially small singularity shield. A positive-density noisy channel with bounded C^2 density is also admissible.

If the predictive law has density g and O is noiseless, the posterior at $Y = y$ is the coarea current

$$d\Pi_y(z) = \frac{g(z)J_O(z)^{-1} d\mathcal{H}^{D-r}|_{O^{-1}(y)}(z)}{\int_{O^{-1}(y)} g J_O^{-1} d\mathcal{H}^{D-r}}.$$

Its factorial coordinates are finite Radon measures and its collision traces are intersections of the observation level set with incoming flux faces. Let $\|\Pi\|_{\text{pc}}$ be the sum of their weighted masses and tangential BV currents over all codimensions generated by observations and collisions.

Lemma 1.1 (Coarea posterior-current stability). *For every admissible observation channel, the Bayes posterior is an element of the posterior-current hierarchy. On a compact control set and outside a superexponentially small set of observations,*

$$\|\Pi_{k+1}\|_{\text{pc}} \leq C \|\Pi_k^{\text{pred}}\|_{\text{pc}}$$

with a constant uniform in the observed value. For a positive-density channel the posterior is the codimension-zero case. The B2 measure-trace renewal semigroup acts on every such current.

Proof. The coarea theorem gives the displayed regular conditional probability and finite mass. Transversality $J_O \geq j_*$ controls its density with respect to Hausdorff measure. Intersecting a C^2 observation level with a regular collision face gives the corresponding incoming flux current; tangential BV is controlled by the second fundamental forms and the predictive trace norm. The bad nontransverse set is contained in the B2 grazing/singularity shield. The B2 renewal operator was constructed on finite boundary measures and hence propagates these currents without requiring an interior density. $\square$

1.2. Prediction followed by observed disintegration. For a block control u , let $\mathcal{P}_{k,u}^\varepsilon$ be the exact predictive pushforward/Feynman–Kac operator on posterior currents. Let $\mathcal{D}_{k,u}^\varepsilon(\pi, y)$ be the coarea or density disintegration kernel. The controlled Bayes update is

$$\Pi_{k+1}^\varepsilon = \mathcal{D}_{k,u_k}^\varepsilon(\mathcal{P}_{k,u_k}^\varepsilon \Pi_k^\varepsilon, Y_{k+1}).$$

This conditions on the new observation and uses the controlled posterior from the preceding blocks.

Theorem 1.2 (Feller belief transition). *On every compact posterior-current ball and compact relaxed-control set, the joint kernel of (Y_{k+1}, Π_{k+1}) is Feller in the weak current topology. It has an analytic graph and admits measurable minimax selectors for finite block games. Ionescu–Tulcea therefore constructs one strategy-dependent law for every nonanticipative relaxed strategy pair.*

Proof. Predictive continuity is the B2 measure-trace semigroup continuity. For a noiseless channel, coarea integrals depend continuously on the current, control, and regular value; the singular values have superexponentially small total mass and are removed by a common compact exhaustion. For a noisy channel this is ordinary dominated convergence. Compact relaxed controls and the Feller kernel give lower/upper semicontinuity of the one-step value. The measurable maximum theorem and the analytic-graph selection theorem yield selectors. $\square$

1.3. Quenched conditional block pressure. Let H be a bounded particle/contact source on one block. For a reachable posterior current π define

$$Q_{k,\varepsilon}^{\pi,u}(H) = \mu_\varepsilon^{-1} \log \int e^{\mu_\varepsilon H} d\mathcal{P}_{k,u}^\varepsilon \pi.$$

The B2 operator proof is linear in the initial finite current, and its constants depend only on the posterior-current ball.

Theorem 1.3 (Strategy-uniform quenched block theorem). *Uniformly over reachable posterior currents in a compact ball, regular observations, and relaxed controls,*

$$Q_{k,\varepsilon}^{\pi,u}(H) \longrightarrow Q_k^{\pi,u}(H)$$

with all finite source derivatives. Conditional on the observation, the same statement holds as a ratio theorem: numerator and denominator have uniform coarea asymptotics and their exponentially small exceptional sets are common. Every regular limiting block pair has a unique normalized tilt and an exponential pressure gap outside each projected neighborhood.

Proof. The B2 one-block tree expansion acts on the initial current by duality. Its cyclic gain and measure-trace norm are uniform on the compact posterior ball, so the consistency error is uniform in π, u . For a noiseless observation, insert the coarea delta by Fourier inversion in the observed coordinate. Submersion transversality supplies the uniform local coefficient; the same coarea denominator appears in the untitled law. Their ratio cancels the surface normalization. The singularity shield is uniform over strategies by compactness. Strict projected curvature is the B3 Lax–Milgram covariance, which gives the pressure gap. $\square$

1.4. Belief contraction and asymptotic sufficiency. Let $\mathbf{m}(\pi)$ be the resolved one-particle density/contact mean of a belief current and let $\mathfrak{h}_j(\pi)$ be its connected hierarchy coordinates of order $j \geq 2$. The B2 connected renewal estimate gives, after one kinetic block,

$$\|\mathfrak{h}_j(\Pi_{k+1})\| \leq C\rho^j \mu_\varepsilon^{1-j} + e^{-c\mu_\varepsilon} \|\Pi_k\|_{\text{pc}}.$$

Observation reweighting preserves this estimate on regular coarea fibers.

Theorem 1.4 (Posterior contraction and value sufficiency). *For every fixed number of blocks, finite posterior marginals concentrate exponentially around the unique kinetic hierarchy determined by the resolved density/contact history. Uniformly over admissible strategies,*

$$\sup_{\pi: \mathbf{m}(\pi)=f} |V_k^\varepsilon(h, \pi) - \bar{V}_k^\varepsilon(h, f)| \longrightarrow 0.$$

Thus the higher posterior correlations are asymptotically insufficient for the controlled value, and the limiting dynamic program closes on the resolved kinetic state.

Proof. Apply the pressure gap in Theorem 1.3 to a finite separating family and then use exponential compactness of posterior-current balls. The connected hierarchy estimate makes the tail uniformly negligible. Couple two beliefs with the same resolved mean through the common normalized block tilt; their next-block payoff and transition laws differ by the displayed vanishing hierarchy tail. Backward induction over the fixed number of blocks proves value sufficiency. $\square$

1.5. The controlled kinetic game. The exact finite information state is (h, π) , with the complete observed history and posterior current. The finite game separates one-time preparation, reward-only control, and adaptive law control. The limiting Hamiltonian is the Isaacs envelope of the B2 kinetic Hamiltonian over relaxed controls.

Theorem 1.5 (Posterior dynamic programming and kinetic limit). *The finite belief game has upper and lower values and satisfies the exact block DPP. Under the Isaacs condition these values converge locally uniformly to the unique B4 viscosity solution on the reduced kinetic state. The proof uses Theorems 1.2, 1.3, and 1.4; it does not replace the posterior by a density before proving sufficiency.*

Proof. Feller belief transitions and measurable selectors give the finite DPP. Uniform quenched block Laplace principles identify the relaxed upper and lower generators. Posterior-current compactness gives containment, and asymptotic sufficiency removes higher belief coordinates. B4 comparison identifies the common limit under Isaacs. $\square$

1.6. Testing, LAN, and continuous-prior denominators. Let $Q_\varepsilon(\theta)$ be a finite-dimensional canonical pressure family. Define the exact score

$$\Delta_\varepsilon = \sqrt{\mu_\varepsilon}(X_\varepsilon - DQ_\varepsilon(\theta_0)).$$

For a continuous prior with density r_0 positive at θ_0 , put $B_\varepsilon = B(\theta_0, L/\sqrt{\mu_\varepsilon})$. Uniform LAN on this ball gives, with high probability,

$$\int_{B_\varepsilon} r_0(\theta) L_\theta d\theta \geq c\mu_\varepsilon^{-d/2} L_{\theta_0} e^{-C(1+|\Delta_\varepsilon|^2)}.$$

This is the missing denominator lower bound.

Theorem 1.6 (KL odds, Chernoff error, exact LAN, and BvM). *Typical log posterior odds have the directed KL rate. Expected posterior error has the minimax Chernoff/testing exponent, with the local denominator above. The local likelihood ratio satisfies*

$$\log \frac{dP_{\theta_0+h/\sqrt{\mu}}}{dP_{\theta_0}} = h \cdot \Delta_\varepsilon - \frac{1}{2} h^T \mathcal{I} h + o_P(1)$$

uniformly on compact h -sets. Under prior thickness and identifiability, the rescaled posterior converges in total variation to the Gaussian with information matrix $\mathcal{I}$.

Proof. The B3 exact-centred Gaussian theorem gives the score CLT and uniform derivative bounds give stochastic equicontinuity. Pairwise fractional likelihood moments give the Chernoff numerator estimate. Divide by the integrated local-ball lower bound, not by a nonexistent prior atom. Uniform tests on fixed complements and on the intermediate LAN annulus follow from the pressure gap and the positive B3 covariance. Taylor expansion at the exact finite mean gives LAN, and the usual local Gaussian integration plus the two testing regions gives total-variation BvM. $\square$

Theorem 1.7 (Round-seven C1 closure). *Exact observations are represented by coarea posterior currents, reachable beliefs remain in the B2 measure-trace domain, block pressures satisfy a strategy-uniform quenched ratio theorem, belief transitions are Feller, higher correlations are proved asymptotically insufficient, and continuous prior testing uses an explicit local denominator. Finite LAN centering remains exact.*

Proof. Combine Lemma 1.1 and Theorems 1.2, 1.3, 1.4, 1.5, and 1.6. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-three dependency ledger. Clean compilation and automated structural checks are

necessary verification gates; independent external journal review remains a separate process.

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